

Obesity and Healthcare Spending

DATA 608 - Story 3

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```
library(tidyverse)
library(ggplot2)
library(readxl)
```

Obesity and Healthcare Spending

Research Question

Do differences in per capita healthcare spending correlate with adult obesity prevalence?

Analysis

Comparing the quintiles of healthcare spending per capita to the obesity rates in each state, we see a minor inverse relationship. This trend is not consistent however; the second lowest health care spend has a much higher IQR for obesity compared to both the 1st and 5th quintiles, and a similar exception can be seen with the fourth quantile. Outliers in both the lowest and highest groups are min and max extremes for the population.

In summary, the inverse correlation between healthcare spend and obesity rates exists, but additional statistical analysis and data would likely better explain the relationship.

```
# Load datasets
obesity_data <- readxl::read_xlsx("adult_obesity.xlsx", sheet = 1)

#www.kff.org/state-health-policy-data/state-indicator/health-spending-per-capita

raw_per_capita <- read_csv("raw_data.csv", skip=2, quote="\") |>
  slice(1:52) |>
```

```

mutate(health_spend_per_capita = as.numeric(str_remove(`Health Spending per Capita`, "$")) |>
rename(state = Location) |>
filter(!(state %in% c("United States", "District of Columbia")))) |>
select(state, health_spend_per_capita)

# Combine datasets
combined_data <- obesity_data |>
  inner_join(raw_per_capita, by = c("State" = "state")) |>
  select(State, `Obesity %`, health_spend_per_capita) |>
  mutate(adult_obesity = as.numeric(str_remove(`Obesity %`, "[%]"))) |>
  drop_na()

# Assign states to quintiles based on healthcare spending

combined_data <- combined_data |>
  mutate(spend_quintile = ntile(health_spend_per_capita, 5)) |>
  mutate(spend_quintile = factor(spend_quintile, labels = c("1st", "2nd", "3rd", "4th", "5th")))

# Calculate spending ranges for each quintile
quintile_ranges <- combined_data %>%
  group_by(spend_quintile) %>%
  summarise(
    min_spend = min(health_spend_per_capita, na.rm = TRUE),
    max_spend = max(health_spend_per_capita, na.rm = TRUE),
    label = paste0(spend_quintile, "\n($", round(min_spend), " - $", round(max_spend), ")")
  ) %>%
  arrange(spend_quintile)

# Build box plot visual with spending ranges and points
ggplot(combined_data, aes(x = spend_quintile, y = adult_obesity, fill = spend_quintile)) +
  geom_boxplot(alpha = 0.7) +
  geom_point(width = 0.2, alpha = 0.5, size = 2) +
  scale_fill_viridis_d(option = "D", end = 0.8) +
  scale_x_discrete(labels = setNames(quintile_ranges$label, quintile_ranges$spend_quintile)) +
  labs(
    title = "Adult Obesity Prevalence in U.S. States by Healthcare Spending",
    x = "Healthcare Spending Quintile (Per Capita)",
    y = "State Adult Obesity Prevalence (%)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 12, face = "bold"),

```

```
axis.title.x = element_text(size = 10),
axis.title.y = element_text(size = 10),
legend.position = "none"
)
```

